

llmXive Automated Review of LoopCoder-v2: Only Loop Once for Efficient Test-Time Computation Scaling

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: [qwen.qwen3.5-122b](#)

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper presents a compelling empirical study on the non-monotonic effects of loop counts in Parallel Loop Transformers (PLT). The core claim—that loop count 2 is optimal due to a gain-cost trade-off involving CLP-induced positional mismatch—is well-supported by the internal diagnostics and the macroscopic benchmark results in Table 1. The numbers in the abstract (SWE-bench Verified 43.0 -> 64.4) align perfectly with Table 1.

However, three specific issues regarding factual accuracy and citation integrity require correction:

1. **Benchmark Naming Inconsistency:** The Abstract and Introduction refer to “Multi-SWE” improving from 14.0 to 31.0. Table 1 lists this metric as “SWE-M” (SWE-bench Multilingual). “Multi-SWE” is not a standard benchmark name and appears to be a conflation. The text should be updated to use the exact label “SWE-M” to match the results table.
2. **Compute Claim Verification:** Section 3.1 states training consumed “a total of 1M GPU hours.” For a 7B model on 18T tokens, this figure is orders of magnitude higher than typical estimates. Unless this includes massive failed runs or refers to a different unit, it is likely incorrect. This load-bearing claim must be verified and corrected.
3. **Future-Dated Citations:** The bibliography cites papers with 2026 publication dates (e.g., `vendrell2026memoryefficient`). As these papers cannot exist yet, the years must be corrected to the actual upload year (e.g., 2025) or replaced with existing works. Citing future-dated papers undermines the literature review’s credibility.

Action items.

- **[writing]** Abstract and Intro claim ‘Multi-SWE’ improves 14.0->31.0, but Table 1 lists ‘SWE-M’ (SWE-bench Multilingual) with these values. ‘Multi-SWE’ is undefined. Rename to ‘SWE-M’ or ‘SWE-bench Multilingual’ to match the table.
- **[science]** Section 3.1 claims training consumed ‘1M GPU hours’ for a 7B model on 18T tokens. This is likely inflated or mislabeled (e.g., TPU hours). Verify the actual compute cost and correct the number or unit to ensure reproducibility.
- **[writing]** Bibliography cites multiple papers with 2026 dates (e.g., `vendrell2026memoryefficient`, `schwethelm2026how`). These cannot exist yet. Correct the years to the actual upload year (e.g., 2025) or replace with existing verified references.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The figure is visually clear and the data trends are easy to interpret. However, the caption contains LaTeX rendering errors for variable names and uses inconsistent variable notation (r vs t) compared to the plot axes.

Figure 2

The figure effectively illustrates the ‘gain-cost scissors’ concept with clear annotations, but it fails to render the 95% confidence intervals mentioned in the caption and contains missing variable symbols in both axis labels.

Figure 3

The figure effectively visualizes the increasing redundancy of attention heads across loops with clear axes and a color scale, but the caption contains a typo (‘Head\$head’) and the colorbar is missing a descriptive label.

Figure 4

The figure is visually clear and supports the caption’s claims regarding attention freezing and gate weights, but there is a variable mismatch where the caption uses ‘r’ while the axes use ‘t’.

Figure 5

The figure is generally readable, but the middle panel’s y-axis is linear despite the caption claiming a log scale, and the caption’s variable notation (r) conflicts with the axis labels (t).

Figure 6

The figure is visually clear with well-labeled bars and axes, but the caption is critically truncated at the end, cutting off the explanation for Loop 1.

Figure 7

Figure 7 effectively communicates the key concepts of PLT loop-count selection, but has minor issues with axis labeling and diagram clarity that could be improved for better understanding.

Action items.

- **[writing]** Figure 1: The caption uses LaTeX placeholders (e.g., $\hat{r}$, $\hat{FP}(r)$) instead of the rendered variable names (e.g., $\Delta_{FP}^{(t)}$, $\Delta_{FP}^{(t)}$) found on the plot axes, creating a disconnect between the text and the visual labels.
- **[writing]** Figure 1: The caption defines the x-axis as ‘loop index r ’, but the plot axes explicitly label the x-axis as ‘Loop index t ’; the variable name should be consistent.

- **[writing]** Figure 1: The caption states ‘Lines: PLT\$_2\$/PLT\$_3\$/PLT\$_4\$’, but the legend explicitly labels these as ‘PLT\$_3\$ (trained \$r=3\$)’, ‘PLT\$_2\$ (trained \$r=2\$)’, and ‘PLT\$_4\$ (trained \$r=4\$)’, the caption should clarify that the subscript denotes the training configuration.
- **[science]** Figure 2: The caption states ‘shaded bands are 95% CIs’, but the rendered figure shows no shaded bands around the data points.
- **[writing]** Figure 2: The right y-axis label contains a missing symbol/variable name (rendered as $\hat{r}$) instead of the intended $\Omega(t)$.
- **[writing]** Figure 2: The left y-axis label contains a missing symbol/variable name (rendered as $\hat{r}$) instead of the intended $\Delta p(t)$.
- **[writing]** Figure 3: The caption contains a rendering artifact ‘Head\$\$head’ which should be corrected to ‘Head–Head’ or ‘Head-to-Head’.
- **[writing]** Figure 3: The colorbar lacks a descriptive label (e.g., ‘Cosine Similarity’) to explicitly define the metric mapped to the color scale.
- **[writing]** Figure 4: The caption defines the x-axis as ‘loop index r ’, but the plot axes are explicitly labeled ‘Loop index t ’, creating a variable mismatch.
- **[writing]** Figure 4: The caption describes the right panel as ‘mean G-SWA gate’, but the y-axis label uses the symbol $\bar{g}(t)$ (implying a mean) without explicitly defining the bar notation in the text.
- **[science]** Figure 5: The caption claims the middle panel is on a ‘log scale’, but the y-axis labels (1.5, 2, 3, 5, 10, 20, 30) are spaced linearly, not logarithmically. This misrepresents the data visualization.
- **[writing]** Figure 5: The caption refers to the middle panel as ‘inter-loop output KL divergence $p(r)$ ’, but the axis label uses the symbol $\Delta p(t)$ and the x-axis is labeled ‘Loop index t ’. The caption variables (r) and axis variables (t) are inconsistent.
- **[fatal]** Figure 6: The caption text is truncated mid-sentence at the end (‘(Loop 1 builds context and is’), leaving the definition of Loop 1 incomplete.
- **[science]** Figure 6: The x-axis label ‘Share of post-context refinement (%)’ and the 0-100 scale imply the bars represent the total distribution of refinement, yet the caption states the bars split the refinement of loops $r \geq 2$. If Loop 1 (context building) is excluded from the ‘post-context’ definition, the bars summing to 100% is correct, but the visual presentation lacks a ‘Loop 1’ segment or explicit note that the 0-100% range applies only to the refinement phase, which could be misleading without the c
- **[writing]** Figure 7: The middle panel’s x-axis is labeled ‘Relative Magnitude’ but the caption describes the trade-off as a function of ‘loop count’ or ‘added loop’; the axis label should explicitly indicate the loop index (r) to match the caption’s description of the gain-cost trade-off.

- **[writing]** Figure 7: The right panel’s ‘Hidden-state Update’ row uses a ‘Start (Loop 1)’ label for the initial point, but the caption refers to ‘Loop 2’ and ‘Loop 3’ as the comparison points; the diagram should clarify that the trajectory begins at Loop 1 and ends at Loop 2/3 to avoid ambiguity about the starting state.
- **[writing]** Figure 7: The right panel’s ‘Representation Diversity’ row uses arrows to indicate ‘Diversity Increases’ and ‘Diversity Decrease’, but the visual representation (scatter plots) lacks a clear legend or color coding to distinguish between the different token types or clusters, making it difficult to interpret the diversity change without external context.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_jargon_police.md

The paper is generally well-written and avoids excessive in-group slang, but it stumbles on the precise definition of its own notation in the mathematical sections. A competent reader from an adjacent field (e.g., a standard NLP researcher not specializing in looped architectures) would likely stall on the definition of the “intrinsic offset cost” $\Omega(r)$ in Section 3.1. The text introduces the concept and immediately presents the equation, but the summation index i and its bounds relative to the sequence length S are not explicitly stated in the prose, forcing the reader to infer the range.

Similarly, in Section 3.2, the description of attention entropy (Eq. 5) contains a grammatical break (“where quantifies whether...”) that obscures the definition of the attention matrix $A^{(r,h)}_{qk}$. While the KL divergence formula (Eq. 6) is standard, the text assumes the reader knows $A^{(r,h)}_q$ is a probability distribution over keys without explicitly stating it. These are minor omissions but represent the exact type of “undefined at first use” barrier this lens polices.

Additionally, the term “Logit Lens rank” in Section 3.3 is used as a standard metric without a brief gloss. While “Logit Lens” is a known concept in the subfield, the specific operationalization here (ranking the ground-truth token) is a specific choice that benefits from a one-clause explanation for a broader audience. Finally, the symbol block is defined in the preamble but used frequently in the body; a brief reminder at its first major appearance in the text would improve self-containment. These issues are easily fixable with parenthetical expansions or minor sentence repairs.

Action items.

- **[writing]** The paper is generally well-written and avoids excessive in-group slang, but it stumbles on the precise definition of its own notation in the mathematical sections. A competent reader from an adjacent field (e.g., a standard NLP researcher not specializing in looped architectures) would likely stall on the definition of the “intrinsic offset cost”

$\sum_{r=1}^R$ in Section 3.1. The text introduces the concept and immediately presents the equation, but the summation index i and its bounds relat

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The paper presents a coherent argument structure: it defines a problem (PLT loop-count selection), proposes a theoretical framework (gain-cost trade-off), and validates it with empirical data and interpretability analysis. The logical flow from the definition of the CLP offset cost to the explanation of the non-monotonic performance curve is sound. The premises (CLP introduces a fixed cost, refinement gains diminish) logically entail the conclusion (performance peaks at loop 2).

However, there are minor logical inconsistencies and presentation gaps that require clarification:

- Data Mismatch (Section 5.1 vs. Table 1):** The text in Section 5.1 states the model attains “33.4% on the agentic SWE-bench-CC.” Table 1 (Section 4.1) reports “31.0%” for “SWE-M” (SWE-bench Multilingual) and does not contain a column or row for “SWE-bench-CC.” This creates a disconnect between the textual claim and the presented evidence. It is unclear if “SWE-bench-CC” is a typo for “SWE-M” (in which case the number is wrong) or a distinct benchmark omitted from the table. The argument relies on these specific numbers to demonstrate the “broad gains,” so the discrepancy must be resolved to ensure the conclusion is supported by the provided data.
- Redundant Phrasing (Section 4.1):** The sentence “In total, training ... consumed a total of 1M GPU hours” contains a tautology (“In total” ... “a total of”). While not a logical fallacy, it degrades the precision of the statement.
- Incomplete Definition (Section 4.2):** Equation 5 defines attention entropy but the subsequent clause “where quantifies whether...” lacks a subject (e.g., “which quantifies”). This breaks the logical link between the formula and its interpretation, leaving the reader to infer the missing subject.

These issues are primarily presentational or require minor data reconciliation rather than indicating a fundamental flaw in the paper’s core reasoning. The central argument—that the fixed cost of the CLP offset eventually outweighs diminishing refinement gains—remains logically valid based on the provided definitions and trends.

Action items.

- [writing]** In Section 4.1 (Table 1), the text states ‘In total, training ... consumed a total of 1M GPU hours.’ The phrase ‘In total’ and ‘a total of’ are redundant. Rephrase to ‘training ... consumed 1M GPU hours’ for clarity.

- **[writing]** In Section 4.2, the definition of attention entropy (Eq 5) ends with ‘where quantifies whether...’, missing the subject (e.g., ‘which quantifies’). This is a grammatical error that obscures the definition’s function. Add the missing subject.
- **[writing]** In Section 5.1, the text claims ‘The same configuration also attains 33.4% on the agentic SWE-bench-CC’. However, Table 1 (Section 4.1) lists ‘SWE-M’ (SWE-bench Multilingual) with a score of 31.0% for the loop=2 model, and does not list ‘SWE-bench-CC’. The text either misnames the benchmark or cites a result not present in the main results table. Clarify the benchmark name and ensure the value matches the reported data or add the missing table entry.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper presents a well-supported mechanistic explanation for the non-monotonic performance of Parallel Loop Transformers (PLT), specifically linking the saturation at two loops to a fixed positional mismatch cost versus diminishing refinement gains. The internal diagnostics (effective rank, offset cost) are rigorously derived from the model’s own states and align with the observed benchmark results.

However, the framing in the abstract, conclusion, and discussion occasionally overgeneralizes findings that are strictly limited to the specific 7B PLT architecture and the specific benchmark suite used.

1. **Scope of “Broad Gains”:** The abstract states the method “delivers broad gains... across code generation, code reasoning, agentic software engineering, and tool-use benchmarks.” While the model performs well on the 10 listed benchmarks in Table 1, the term “broad” implies a universality across the entire domain of code and reasoning tasks that was not empirically tested. The evidence is strictly limited to the specific suite of 10 benchmarks. This rhetorical inflation should be corrected to “gains across the evaluated benchmarks” to accurately reflect the evidence.
2. **Generalizability of the Diagnostic:** The conclusion asserts that the findings provide “interpretability-grounded diagnostics for loop-count selection without exhaustive benchmark sweeps.” This implies the proposed diagnostic (monitoring effective rank) is a general tool applicable to any looped model or architecture. The paper only validates this diagnostic on a single 7B PLT model family. There is no evidence that this specific heuristic (peak at loop 2) holds for other model families, different attention mechanisms, or larger scales. The claim should be scoped to “diagnostics for PLT loop-count selection” to match the demonstrated scope.

3. **Universality of Complementarity:** In Section 5.2, the authors conclude that latent recurrence and explicit reasoning are “complementary axes of test-time computation.” This is a strong, universal claim derived from a single experiment on a 7B model where the combination yielded super-additive gains. While the result is significant, asserting this as a general property of test-time compute rather than a property of this specific model configuration overreaches the evidence. The language should be hedged to reflect that this complementarity was “observed in our setup.”

These are primarily issues of rhetorical precision. Narrowing the scope of the general claims to match the specific experimental boundaries will resolve the overreach without requiring new experiments.

Action items.

- **[writing]** Abstract claims ‘broad gains... across code generation, reasoning, agentic, and tool-use.’ Table 1 shows only 10 specific benchmarks. Replace ‘broad gains’ with ‘gains across evaluated benchmarks’ to match the specific evidence scope.
- **[writing]** Conclusion claims findings provide ‘diagnostics for loop-count selection’ generally. Evidence is limited to one 7B PLT model. Narrow to ‘diagnostics for PLT loop-count selection’ or add cross-architecture validation.
- **[writing]** Section 5.2 concludes latent recurrence and explicit CoT are ‘complementary axes’ universally. Evidence is from one 7B model config. Rephrase to ‘suggests complementarity in our setup’ to avoid overgeneralization.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

This paper presents a methodological study on the efficiency and performance trade-offs of Parallel Loop Transformers (PLT) for code generation. The work involves training and evaluating model variants on standard public benchmarks (e.g., SWE-bench, HumanEval+, MultiPL-E) using a pretraining corpus described as an “internal deduplicated mixture.”

From a safety and ethics perspective, the paper does not present foreseeable, non-trivial risks that are unaddressed:

1. **Dual-Use:** While the model is a code generator, the paper focuses on architectural efficiency (loop counts) and interpretability. It does not introduce novel capabilities for generating malware, exploits, or disinformation, nor does it lower the barrier to such tasks in a way that requires specific mitigation beyond standard model release practices.
2. **Data Privacy & Consent:** The training data is described as an internal, deduplicated mixture. The paper does not release the raw training data, nor does it contain examples of

Personally Identifiable Information (PII) in the text or figures. The evaluation uses standard public benchmarks. No human-subjects research (surveys, interviews) is conducted; thus, IRB/consent statements are not required.

3. **Bias & Fairness:** The paper reports performance across various benchmarks but does not explicitly surface or analyze demographic bias (e.g., performance disparities across languages or cultural contexts) in a way that suggests a hidden harm. While a deeper bias analysis could be valuable, the absence of a specific, identified harm to an identifiable group in the current results does not constitute a failure to disclose a known risk.
4. **Vulnerability Disclosure:** The paper does not report discovering or exploiting security vulnerabilities in live systems; it evaluates model performance on static benchmarks.

The research is a standard algorithmic efficiency study. No specific safety disclosures or mitigations are missing that would prevent publication.

Scientific Evidence

Weights the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The paper presents a compelling empirical study on the non-monotonic effects of loop counts in Parallel Loop Transformers (PLT), supported by a large-scale training effort (18T tokens, 1M GPU hours). The macroscopic results in Table 1 clearly show a performance peak at loop count 2, followed by degradation at loops 3 and 4. The microscopic analysis (Figures 1-6) provides a plausible mechanistic explanation involving the trade-off between representational refinement and the fixed cost of the Cross-Loop Position (CLP) offset.

However, the evidentiary strength of the central claim—that the CLP offset specifically causes the saturation—is weakened by two design gaps. First, the results for both the benchmark performance (Table 1) and the internal diagnostics (Figures 1-6) are presented as single-point estimates or mean curves without any reported variance (standard deviation, standard error, or confidence intervals) across random seeds. In deep learning, especially with large models and complex recurrent dynamics, performance can vary significantly between seeds. The claim that loop-3 “regresses” (e.g., SWE-bench dropping from 64.4 to 27.6) is a strong assertion that requires demonstrating this drop is consistent across re-initializations, not a lucky seed for loop-2 or an unlucky one for loop-3. Without error bars or seed counts, the reader cannot distinguish a robust structural effect from sampling noise.

Second, the attribution of the performance drop to the “CLP offset cost” is not fully isolated. The models with different loop counts differ in their total effective depth (loop count $\times$ layers) and training dynamics. While the authors argue the CLP offset is the culprit, they do not provide an ablation that isolates this specific mechanism. For instance, a comparison between a PLT model with CLP and a variant without CLP (but with the same loop count and effective

depth) would be necessary to prove that the offset, rather than general instability in deeper recurrent unrolls or optimization difficulties, is the primary driver of the degradation. Without this control, the “gain-cost” explanation remains a plausible hypothesis rather than a proven causal mechanism.

To strengthen the evidence, the authors should report results averaged over multiple random seeds (e.g., 3-5) with error bars for both the benchmark scores and the key diagnostic metrics (effective rank, offset cost). Additionally, an ablation study isolating the CLP mechanism (e.g., removing the shift while keeping the loop count and architecture fixed) would be required to definitively rule out alternative explanations for the performance regression.

Action items.

- **[writing]** The paper presents a compelling empirical study on the non-monotonic effects of loop counts in Parallel Loop Transformers (PLT), supported by a large-scale training effort (18T tokens, 1M GPU hours). The macroscopic results in Table 1 clearly show a performance peak at loop count 2, followed by degradation at loops 3 and 4. The microscopic analysis (Figures 1-6) provides a plausible mechanistic explanation involving the trade-off between representational refinement and the fixed cost of the Cros

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The paper provides a detailed mechanistic analysis of loop-count dynamics, supported by internal diagnostics computed over 500 samples with reported 95% confidence intervals (Figures 1-5). However, the statistical treatment of the primary macroscopic results in Table 1 is incomplete.

Table 1 presents benchmark scores (e.g., 64.4% for Loop=2 on SWE-bench) as single point estimates. While the internal figures explicitly state “500 samples” and show confidence intervals, the text does not specify the number of independent training seeds used to generate these benchmark scores, nor does it report the standard deviation or confidence intervals for these values. In deep learning, performance can vary significantly across random seeds. Reporting a single number without uncertainty quantification makes it impossible to distinguish robust signal from run-to-run noise. The authors should report the mean and standard deviation (or 95% CI) across at least 3 independent training runs for all benchmark results.

Additionally, the text uses terms like “significantly better” (Abstract; Section 4.1) without citing a formal statistical hypothesis test, p-value, or effect size. If no test was performed, the language should be softened to “substantially better” or “notably better,” and the absolute difference with its uncertainty should be provided. These are reporting issues that can be resolved by aggregating existing run data or clarifying the statistical basis of the claims.

Action items.

- **[writing]** Table 1 reports benchmark scores as single point estimates without uncertainty. The text mentions 500 samples for internal diagnostics but not for benchmark results. Report mean $\pm$ SD or 95% CI across 3 training seeds for all benchmark scores to quantify run-to-run variance.
- **[writing]** The abstract and Section 4.1 claim the model is ‘significantly better’ without reporting a statistical test, p-value, or effect size. Replace ‘significantly’ with ‘substantially’ or report the specific test used (e.g., paired t-test) and its p-value to support the claim.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The paper is generally well-structured and the core argument regarding the gain-cost trade-off in Parallel Loop Transformers is presented clearly. However, there are several specific instances where the prose stumbles, requiring re-reading or causing momentary confusion.

First, there are minor grammatical slips and redundancies. In Section 3.1, the phrase “consumed a total of” is tautological. In Section 3.2, a sentence defining the attention entropy metric is missing its subject (“where quantifies...”), rendering it a fragment. These are easily fixed but disrupt the flow.

Second, the paper suffers from some structural redundancy. The “Discussion” section (Section 5) largely restates the findings of the “Synthesis” subsection in Section 4.2 with nearly identical headings and content. A Discussion should ideally interpret the *implications* of the findings rather than re-summarize the data points already presented in the Results. Merging these or reframing the Discussion to focus on broader implications would improve the narrative arc.

Third, there are issues with signposting and definition. In Section 4.1, the text mentions “SWE-bench-CC” and a score of 33.4%, but this benchmark is not listed in the main results table (Table 1) nor defined in the text. The reader is forced to guess what this metric is or where the number comes from. All metrics should be introduced or referenced clearly upon first mention.

Finally, the transition between the definition of the “Intrinsic offset cost” and its computability in Section 3.2 is slightly clunky, breaking the logical flow of the definition. Tightening this sentence structure would make the mathematical contribution clearer.

Addressing these points will ensure the reader can move through the paper without having to pause to parse awkward phrasing or hunt for missing definitions.

Action items.

- **[writing]** Section 3.1, ‘Training Protocol’: The sentence ‘In total, training LoopCoder-v2 of different loops in this work consumed a total of 1M GPU hours’ is redundant (‘total... a total of’). Also, the phrase ‘LoopCoder-v2 of different loops’ is awkward; rephrase to ‘training the

LoopCoder-v2 variants with different loop counts’.

- **[writing]** Section 3.2, ‘Per-Loop Hidden-State Dynamics’: The paragraph defining ‘Intrinsic offset cost’ contains a sentence fragment: ‘This per-loop scalar $\Omega(r)$ is computable directly from the neighboring hidden states of the LLM.’ This sentence is grammatically complete but feels disconnected from the preceding definition. Better: ‘We define this per-loop scalar $\Omega(r)$ as... which is computable directly from...’
- **[writing]** Section 4.1, ‘Main Results’: The sentence ‘The same configuration also attains 33.4% on the agentic SWE-bench-CC, confirming that the loop-2 gains carry over to held-out agentic settings’ appears abruptly. The benchmark ‘SWE-bench-CC’ was not introduced in the text or table caption (Table 1 lists ‘SWE’ and ‘SWE-M’). Define the acronym or refer to the table entry clearly.
- **[writing]** Section 5, ‘Discussion’: The paragraph ‘Loop 2 is the principal site of productive refinement’ repeats the exact heading and content of Section 4.2 (‘Synthesis’). This creates a redundant structure where the Discussion merely restates the Results. Merge the Discussion’s specific findings into the Results synthesis or reframe the Discussion to focus on implications rather than re-summarizing the data.
- **[writing]** Section 3.2, ‘Attention Heat-Map Evolution’: The sentence ‘where quantifies whether a head is globally diffuse or locally focused at loop r ’ is missing a subject. It should read ‘where $H(r,h)_q$ quantifies whether...’ or ‘which quantifies whether...’.